Training-Consistent Pressure-Prior Integration for Independent Multi-Agent Traffic Signal Control: Confirmatory Evidence and Setting-Dependent Boundaries

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Abstract

Pressure is the organizing quantity in classical traffic signal control and several reinforcement learning (RL) formulations. However, in the published pressure-informed RL controllers of which we are aware, pressure enters mainly through the reward or the observation of value-based agents. This study evaluates a different form of integration: a pressure-based action prior for independent proximal policy optimization (PPO) agents. The prior represents a per-intersection preference for signal phases that relieve queue imbalance, is added to the policy logits, and is applied consistently during rollout sampling and PPO updates. Under an internally pre-registered, hash-locked protocol frozen before data collection, the resulting method, IPPO-P, was compared with a matched IPPO baseline. The evaluation covered three synthetic grid scenarios spanning two network topologies: a 3×3 grid and a 2×2 grid at two demand levels. It used ten new independent training seeds, five deterministic evaluation episodes per task in fresh environments, and one pre-registered primary outcome: demand service time (U; lower is better). U is the mean per-vehicle service time accrued by all planned demand vehicles, with unfinished vehicles censored at the demand horizon. All 60 planned tasks were completed, the demand-state bookkeeping check was satisfied, and no teleport events occurred across the 300 evaluation episodes. IPPO-P improved U by 1.712% (seed-level log-ratio effect; descriptive bootstrap 95% CI [1.252%, 2.202%], 20,000 paired resamples). The equal-weight composite covers two topology families, and H2 and H3 share the 2×2 topology, so two-thirds of the composite weight is assigned to that topology family. All ten training-seed effects were positive, with an exact two-sided paired sign-flip $p = 0.001953$, corresponding to a descriptive range of approximately 0.5–4.5 s of service time per vehicle across scenarios. Scenario-specific effects (+0.407%, +1.023%, +3.675%) were secondary and descriptive. Post-confirmation exploratory scale-ups showed that the gain was not universal. The effect was positive at 2700 veh/h (+0.458%, $n = 5$), negative at 5400 veh/h (−0.997%, 0/5 seeds positive, $n = 5$), and positive in direction but underpowered at 6300 veh/h ($n = 2$, one near-tie). Descriptive non-learning anchors bound the scale of the comparisons. Both learned policies performed well below a fixed-time program, and IPPO-P's mean U was 2.9%–6.9% lower than that of a constrained descriptive max-pressure variant with a 33 s minimum green and halting-count proxy, which in turn edged IPPO in the heaviest scenario. These figures provide scale context rather than ranked claims. Within this IPPO-internal controlled evaluation at a single frozen prior scale, training-consistent pressure-prior integration improved demand service time relative to matched IPPO, while setting-dependent boundaries caution against extrapolation.

Index Terms

traffic signal control; multi-agent reinforcement learning; independent proximal policy optimization; action prior; internally pre-registered confirmatory evaluation; demand service time

A. Introduction

Traffic signal control (TSC) is a sequential decision problem in which locally acting controllers jointly determine network-level delay. Each controller's best action depends on queue states that neighboring controllers are simultaneously shaping, so independent adaptation makes the environment non-stationary from each agent's perspective. Stable policy learning under this coupling, through explicit coordination, architectural inductive bias, or training-stability safeguards, is therefore a central concern. Existing practice spans fixed-time plans, actuated rules, and learning-based controllers [1]. Multi-agent reinforcement learning (MARL) entered the field early [2]. Deep RL subsequently produced diagnostic single-intersection agents [3], phase-competition representations [4], parameter-shared decentralized policies for large networks [5], and graph-based network-level cooperation [6]. These mechanisms balance decentralization and coordination assumptions in different ways. Across these learning-based designs, policies are trained from reward signals. The learning process converts traffic state into behavior without an explicit structured preference over actions.

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A parallel line of work assigns a principled role to pressure. Backpressure scheduling stabilizes constrained queueing networks with maximum-throughput guarantees [7]. Max-pressure control extends this principle to signalized intersection networks with well-characterized stability properties [8]. In deep RL for TSC, pressure has mainly served as a learning signal. PressLight trains agents to minimize network pressure through the reward [9]. The PRCOL metric introduced in PDLight incorporates downstream lane capacity into the pressure definition and is used in the reward [10]. Multi-hop upstream pressure extends the quantity beyond adjacent links and is used in both the observation and reward [11]. Reported pressure-informed RL controllers in this line are largely DQN-family agents that place pressure in the reward or state. The pressure signal does not act directly on the sampling distribution of an on-policy policy.

Proximal policy optimization (PPO) [12] is a widely used on-policy algorithm with clipped-ratio updates. Independent PPO (IPPO), in which each agent learns from local observations without communication or a shared critic, performed unexpectedly well in the StarCraft multi-agent challenge [13]. MAPPO, by contrast, demonstrated the value of centralized value functions in cooperative games [14]. For TSC, IPPO matches the deployment-relevant setting of decentralized execution from local observations. However, its policy update does not represent the traffic-engineering preference encoded by pressure. Nothing in the PPO objective distinguishes, a priori, phases that relieve queue imbalance from those that do not. Whether a pressure-derived prior can improve service quality when integrated consistently into the policy distribution during training remains an empirical question.

The empirical TSC-RL literature has grown rapidly [15], but reported gains are sensitive to protocol choices and random-seed variation [16], [17], [18]. Deep RL results vary across training runs, data orders, and other stochastic sources [16], [18]. Small numbers of runs can therefore produce unstable effect estimates and conclusions [17], limiting the evidential value of single-seed or single-scenario comparisons. This paper is deliberately structured as a confirmatory evaluation. The protocol, primary contrast, metric, and analysis were frozen internally under a hash-locked, author-controlled protocol (Section 4.1) before confirmatory data generation. Results are reported with seed-level transparency, including settings in which the intervention does not help.

The intervention studied here is narrow. IPPO-P (pressure-prior IPPO) adds a per-intersection, per-phase pressure vector to the actor logits before categorical sampling. The vector is centered and standardized within each intersection, clipped to $[-3, 3]$, and scaled by 2.25. The clip never binds at the tested four-phase intersections (Section 3.3). The same observation, prior vector, temperature, sampled action, and old log-probability are used during rollout and PPO update. The reward, actor observation vector, network architecture, and training budget match those of the IPPO baseline. However, IPPO-P additionally consumes the halting-count pressure state used to construct the logit prior. The evaluated treatment is therefore the complete prior-integration package, including the extra halting-count channel, rather than an isolated change to logit arithmetic. The contributions are limited to what the current evidence directly supports:

1. a precisely specified, reproducible IPPO-P implementation with rollout/update distribution consistency (stored-prior reuse, specified at tuple level in Section 3.4);
2. an internally pre-registered (hash-locked), seed-level confirmatory comparison against IPPO over three scenarios across two topology families, with H2 and H3 representing two demand levels on the same 2×2 network, using ten new independent training seeds;
3. an explicit report of setting-specific boundary evidence from post-confirmation exploratory extensions, including a negative result at 5400 veh/h and an underpowered probe at 6300 veh/h.

The research question is whether consistently incorporating a pressure-based action prior into rollout action selection and PPO updates improves demand service time over standard IPPO in the tested multi-intersection scenarios. A secondary question is how the observed effect varies across exploratory settings. The confirmatory hypothesis is that IPPO-P reduces demand service time relative to IPPO across H1, H2, and H3, with equal scenario weights, over ten new independent training seeds. By design, this weighting assigns two-thirds of the composite weight to the 2×2 topology family because H2 and H3 use the same network at different demand levels. Exploratory questions concern scenario-specific effect variation, generalization to 3×3 scale-up settings, and, strictly as a hypothesis, a variance-stabilization account of the prior's effect.

This paper proposes no new coordination architecture, pressure metric, or learning algorithm; every component except the logit prior is standard. Its value lies in the quality of the comparison: one frozen protocol, one internally pre-registered contrast, ten paired seeds, integrity audits for every episode, and an explicit exploratory annex that reports negative and positive results with equal care. Whether a domain prior belongs within the training distribution of an on-policy controller cannot be resolved by single-run comparisons. The confirmatory design is intended to address that question.

Section 2 relates this work to prior literature, Section 3 specifies the method, Section 4 the experimental design and statistical analysis, Section 5 reports results, Section 6 discusses findings and limitations, and Section 7 concludes.

B. Related Work

1) Reinforcement learning for traffic signal control: Multi-agent RL was applied to traffic lights well before the deep learning era, with agents learning value functions from local delay feedback [2]. Fixed-time plans and actuated rules remain the deployed defaults against which learning methods are judged [1]. Deep RL subsequently reshaped the field. IntelliLight systematized reward and pipeline design for a single intersection [3]. FRAP represented competition between signal phases as a learnable structure [4]. MPLight introduced parameter sharing, allowing one policy to scale across hundreds of decentralized intersections [5]. CoLight used graph attention over neighboring agents to learn network-level cooperation [6]. These mechanisms occupy different points on the coordination spectrum. Parameter sharing reuses capacity without coupling execution, whereas graph communication couples agents through learned message passing during both training and execution. Surveys organize this design space by state representation, reward construction, and coordination mechanism [1], [15]. Two features motivate the present design. First, evaluation protocols vary widely in simulator choice, demand generation, metric definitions, and random-seed counts, making cross-paper effect comparisons unreliable. Second, decentralized policy learning remains attractive for deployment because it requires neither inter-agent communication nor a centralized coordinator during execution.

2) Independent and multi-agent PPO: PPO optimizes a clipped-surrogate objective through stable on-policy updates [12]. In the StarCraft multi-agent challenge, IPPO used local-observation policies and critics without communication and matched or exceeded cooperative methods [13]. MAPPO showed that a centralized value function can improve learning in cooperative games [14]. Together, these results define a design space. Centralized training with decentralized execution mitigates non-stationarity through a global value function during training, but requires global state and couples agents through the critic. Independent PPO requires neither, but treats co-learning agents as part of the environment. For TSC, the independent end of this spectrum is a conservative choice because intersections act on local observations without an embedded coordination structure. IPPO is therefore a meaningful decentralized baseline, but it lacks a mechanism for expressing traffic-domain preferences. This paper examines whether a domain prior, injected into the action distribution and applied consistently during training and inference, changes measured service quality under matched experimental conditions. Whether IPPO improves with greater capacity or different hyperparameters is outside the scope of this design.

3) Pressure-based control and pressure-informed policies: Pressure represents the imbalance between upstream and downstream queues. It originated in backpressure scheduling for queueing networks, where it yields maximum-throughput stability [7], and was established for signalized networks through max-pressure control [8]. Under specified network and demand conditions, max-pressure control selects the phase with the greatest net pressure drop. Within deep RL, PressLight showed that pressure minimization can serve as a reward for coordinated arterial control [9]. PDLight's PRCOL metric refined the pressure definition by accounting for remaining outgoing-lane capacity and used it in the reward [10]. More recently, multi-hop upstream pressure extended the quantity beyond adjacent links and used it in both the observation and reward for anticipatory control [11]. Across this lineage, pressure progressed from raw queue imbalance [9] to capacity-aware pressure [10] and multi-hop anticipatory pressure [11]. The reported agents are value-based (DQN-type), with pressure shaping rewards or states rather than the sampling distribution of an on-policy policy. This statement characterizes the reviewed literature [9]–[11] and is not an exhaustive field-level claim.

4) Positioning: Four distinctions position the present work. First, classical max-pressure control is a deterministic rule with stability analysis [8], not a learned policy. The post hoc comparison in Section 5.7 provides descriptive context rather than a pre-registered or inferential comparison. Second, pressure-based reward shaping changes the optimization objective [9], [10]. Here, the reward is fixed and identical across methods, so the prior's effect is not confounded by a changed objective. Third, pressure can enter a learned policy network as auxiliary state [11]. Here, the actor observation vector is unchanged, but a separate halting-count channel constructs a prior that acts additively on the policy logits. The distinction between adding pressure to policy logits and supplying it as actor state is structural and is not empirically isolated here. Fourth, biasing actions only during evaluation would alter the sampling distribution without changing training and would break PPO's importance-ratio semantics. The evaluated intervention is instead training-consistent and applies the same modified distribution during rollout and update (Section 3.4). Two additional comparison hazards merit attention. An effect attributed to a "prior" may instead arise from accompanying changes in reward scale, state dimension, or network capacity; this design holds those factors fixed. In on-policy methods, the sampling distribution is part of the training algorithm, so modifying it only at inference changes the evaluated object rather than the trained one. This study evaluates treatment consistency under an internally pre-registered, hash-locked protocol, following established pressure-informed work [9], [10], [11]. It makes no novelty-priority claim over pressure-based RL.

C. Method

1) Problem formulation: Experiments use the SUMO microscopic traffic simulator [19]. The simulation advances internally at 1 s resolution and uses the default passenger vehicle type. The implementation explicitly sets `--time-to-teleport 1000` s rather than SUMO's default 300 s. Thus, the zero-teleport outcome reported in Section 5 remains

an empirical integrity gate rather than a configuration guarantee. Each signalized intersection is controlled by one agent. Its observation concatenates per-lane density over controlled lanes, defined as vehicle count normalized by lane capacity at a 7.5 m minimum gap, with a one-hot encoding of the current green phase. The action space comprises the admissible green-phase indices. A 2 s yellow interval is enforced between conflicting green phases. This interval is shorter than the 3–5 s typical of urban geometry, but is fixed across methods and was not varied. Agents act at 10 s decision intervals. Each episode spans a 10,000 s demand horizon during which vehicles enter at fixed route-flow rates (Table 1). An agent-requested phase change takes effect only after the yellow interval and the current phase have expired. Decision opportunities therefore vary across intersections within a step, making the transition process semi-Markov at the agent level.

The primary outcome is demand service time U, defined across all planned demand vehicles as the mean of max(0, service_end − scheduled_departure). A vehicle that arrives normally within the demand horizon contributes its arrival time, recorded on the 10 s simulation-step grid. An arrival within a step is credited to the closing step boundary and therefore never exceeds the horizon; the horizon cap binds only at the boundary. Every other planned vehicle, whether not yet inserted, still in the network, temporarily out, or abnormally arrived, is censored at the horizon end. Thus, U is not the mean travel time of completed vehicles. Unfinished demand accrues the maximum service time rather than being excluded. This censoring makes U sensitive to how a policy handles demand near the horizon and resistant to completion-count inflation. Such inflation can favor policies that complete easier trips while stranding others. Units are seconds, and lower values are better. The local reward uses tanh-normalized speed, queue, and waiting levels weighted 0.42/0.22/0.36 and directional changes weighted 0.52/0.18/0.30. It also includes a weak action heuristic and congestion and alignment penalties. The global term uses the same current and delta metrics and is mixed with each local reward at weight 0.30. The reward is identical across methods and separate from evaluation; U is not the optimized reward. The exact frozen source artifacts are `environment/traffic_signal.py` and `environment/env.py`, whose hashes are recorded in each `protocol_state.json`; the many lower-level subcoefficients are not reproduced here.

2) IPPO baseline and shared experimental controls: Each agent has an independent actor and critic, both implemented as two-layer multilayer perceptrons with 128 units and tanh activations. Training uses $\gamma = 0.99$, GAE $\lambda = 0.95$, ratio clip 0.20, Adam with learning rate 3×10^{-4}, entropy coefficient 0.01, value coefficient 0.5, five update epochs, and gradient-norm clipping at 0.5. Advantages are computed through generalized advantage estimation from each agent's value function, which is trained on the cooperative reward stream. Episodes end through simulation-time truncation at 10,000 s rather than at a terminal state. The advantage estimate therefore uses a zero next-state value at the truncated final step, corresponding to terminal-style time-limit handling rather than critic-value bootstrapping across truncation. This handling may bias finite-horizon advantage estimates, although it is applied symmetrically across arms and remains a limitation. With $\gamma = 0.99$ over 10 s steps, credit is attenuated by a factor of e over approximately 100 steps, or 1,000 s of simulation time. Per-episode training logs, with one record per training episode, are retained for every task and included in the data deposit. Learning dynamics are therefore reconstructable, although they are not analyzed here. There is no inter-agent communication, parameter sharing, or centralized critic. The IPPO-P critic is structurally identical to the IPPO critic and receives the same local observation. The prior never enters the critic, reward, or observation. IPPO and IPPO-P share the environment configuration, reward, training schedule, random-seed structure, and evaluation procedure. The treatment difference is the pressure-prior pathway, with scale 0.0 for IPPO and 2.25 for IPPO-P, as fixed in the frozen method lock; this pathway includes the separate halting-count channel described below.

3) Pressure prior construction: For intersection i and candidate green phase a, the raw pressure score is the net halting-queue pressure: the sum, over the phase's incoming–outgoing lane pairs, of (halting count on the incoming lane) − (halting count on the outgoing lane), taken from the last simulation step. Raw scores are normalized within each intersection:

~{p}_ia = clip((p_ia − mean_a p_ia) / max(SD_a(p), 1.0), −3, 3),

where the mean and standard deviation are taken over the phases of the same intersection and the standard deviation has a floor of 1.0 (pressures are vehicle counts); the standard deviation is the population SD over the intersection's phases (ddof = 0). Raw halting-queue imbalance is used rather than the capacity-aware refinement of PRCOL [10] because the prior's role is a relative preference among phases at decision time, not a throughput estimate. The IPPO-P policy logits are the baseline actor logits plus $2.25 \times$ ~{p}_ia, with policy temperature 1.0 and no hold bonus for the current phase (the implemented hold-prior machinery carries a zero bonus in the frozen configuration). Three properties of this construction matter (Figure 1). Centering within an intersection removes the overall congestion level, so the prior encodes the relative pressure preference among phases rather than how busy the intersection is. Scaling by the within-intersection standard deviation (floor 1, because raw pressures are small-integer counts) keeps the logit perturbation commensurate across intersections and demand regimes without hand-tuned per-scenario constants. Clipping to [−3, 3] is an implementation safeguard that never activates in the configurations studied here: the standardization is taken over the n = 4 green phases of one intersection under the population-SD convention, so the maximum attainable standardized magnitude is $\sqrt{(n-1)} \approx 1.73$, and the effective worst-case logit shift is $\pm 2.25 \times \sqrt{3} \approx \pm 3.90$. We therefore

do not attribute any binding role to the clip; a sensitivity protocol over the clip range and the scale remains future work (Section 6.7). One sensing asymmetry should be recorded: the prior consumes per-lane halting counts, a channel that is not part of the observation vector (which contains lane densities and the phase one-hot). Both actors receive the identical observation, but IPPO-P's action distribution depends on strictly more state information than IPPO's; the comparison evaluates the prior integration as a whole, this channel included.

4) Training-consistent PPO likelihood: The likelihood specification is given at tuple level (Figure 1). The on-policy buffer is appended at every environment step (one tuple per agent per 10 s step) and consumed by a PPO update after each episode; no decision-slot filtering is applied. Each tuple records the observation, the sampled action, the rollout log-probability under the prior-augmented distribution, the critic value, the step reward, the done flag, and the prior vector. All outer 10 s environment steps enter the buffer, including steps without a new phase decision because the current green has not expired, as well as steps that initiate a yellow transition. A newly initiated 2 s yellow transition is completed internally within that same outer step before simulation advances to its 10 s boundary; it is not a separate sampled buffer step. The tuple retains the agent's sampled action and rollout log-probability even when the environment keeps the current phase or executes the transition, so executed-action statistics differ from sampled-action statistics by construction. During the PPO update, the new log-probability of the stored action is recomputed from the stored observation, and the stored prior vector is reused verbatim; there is no recomputation of the prior from environment state at update time. The update therefore evaluates exactly the same prior-augmented distribution family that generated the data, and the importance ratio exp(log π_new − log π_old) compares two members of that family. Scale and temperature are applied in the same order in both places (logits + 2.25 · ~{p}, then division by the temperature, 1.0 throughout), and the clip was applied once at prior construction, so rollout and update share the same clipped prior. Advantages are computed per episode with $\gamma = 0.99$ and $\lambda = 0.95$, using a zero next-state value at the truncated final step and standardized per agent within the update. Because the stored prior is a deterministic function of the halting state at collection time and is itself stored, the modified policy remains a valid conditional distribution over phases at every stored tuple, and PPO's entropy bonus and clipped-ratio objective operate on it unchanged. The actor and critic are unchanged; the prior is an additive, input-dependent shift of the logits. No decision-slot fraction was archived, so effective sample size by executable decision opportunity is not quantified.

The consistency requirement defines the treatment. Biasing actions only during evaluation would make the behavior policy differ from the policy that generated the stored old log-probabilities. This difference would invalidate the clipped importance ratio and change the quantity estimated by PPO. The prior also shapes the states visited by the fleet during the 50 training episodes, so its influence extends beyond evaluation-time logit arithmetic. The comparison with IPPO therefore evaluates the complete training-consistent prior integration rather than an isolated inference-time bias.

5) Training and evaluation protocol: Each task trains one method on one scenario with one training seed for 50 episodes. A single checkpoint is written after the final episode, containing the final actor and critic weights and configuration. Optimizer state is retained only in the per-episode resume files. Evaluation therefore uses the end-of-training policy without best-episode selection. The checkpoint is reloaded in fresh environments with learning disabled and deterministic action selection. IPPO-P selects the argmax of the same modified logits, whereas IPPO selects the argmax of the baseline logits. Thus, the deployed control law is the trained prior-modified policy rather than a stochastic sampler. Stochastic-sampling evaluation was not assessed; deterministic selection is applied symmetrically to both methods. Five evaluation episodes use fixed evaluation seeds 1042–1046. These seeds affect evaluation only through reseeding the Python, NumPy, and PyTorch generators and the simulator's seed for vehicle insertion and internal randomness. Because actions are selected by argmax, each evaluation episode is a deterministic function of its seed and checkpoint, and evaluation seeds do not affect action sampling. Scenario bindings are listed in Table 1. Evaluation settings are identical across tasks except for scenario binding and seed. The primary confirmatory population comprises ten new independent training seeds, 47–56. Pilot training seeds 42–46 are excluded from primary inference (Section 4.1).

6) Outcome measures: The primary outcome is demand service time U as defined in Section 3.1. Secondary service metrics logged over the same episodes are: mean travel time and delay of completed vehicles; lane-aggregated queue length (mean halting count × 7.5 m over unique controlled lanes), waiting time (mean accumulated waiting over unique vehicles on those lanes), and speed (vehicle-weighted mean over the lane union). These secondary metrics are computed over the same evaluation episodes as U, which makes them correlated with the primary outcome rather than independent replications of it. Only U and the operational audits were pre-specified for reporting (Section 4.5); the secondary service metrics were logged but excluded from the pre-registered analyses and are not reported or interpreted in this paper; no descriptive or inferential claim, and no scenario-specific test, is based on them. Completion bookkeeping retains two distinct denominators: planned_completion_rate = normal compliant arrivals / planned demand, and completion_rate = completed vehicles / departed vehicles. In the 300 confirmatory evaluation episodes the two coincide numerically, but they answer different questions (demand fulfillment versus fleet-level throughput) and would diverge whenever insertions fail or abnormal arrivals occur. The integrity audits have asymmetric epistemic status and are reported as such: the demand-conservation check is a bookkeeping identity (each planned vehicle is assigned exactly one demand

state) satisfied by construction of the accounting routine, and it is not an independent physical-flow assertion (arrived + in-network = departed is not separately tested); the substantive hard gate is the zero-teleport requirement, which is enforced at aggregation and would invalidate the metric interpretation if violated. Completion-rate statistics provide the operational evidence that complements these checks.

D. Experimental Design and Statistical Analysis

1) Pre-registration and separation of pilot from confirmation: The confirmatory protocol (`ippo_prior_confirmation_v1`) was frozen on 2026-10-24 at 22:46:37+08:00, before any confirmatory data were generated. The earliest confirmation task started at 2026-10-24 22:48:16, 1 minute 39 seconds after `frozen_at`, as recorded in `H1_IPPO_seed47/protocol_state.json`. The registration is internal rather than registry-based: the frozen protocol file and its SHA-256 hash, together with the 2026-10-24 freeze timestamp, are author-controlled artifacts, and no external registry entry (e.g., OSF or AsPredicted) exists; the term "pre-registered" is used in this internal, hash-locked sense throughout the paper. The frozen file enumerates the training seeds, evaluation seeds, scenario bindings, hypotheses, the primary metric with its censoring rule, the primary contrast and test, and the method lock (prior scale 2.25, temperature 1.0, configuration `pressure_zscore_hold`); the remaining lock-referenced constants (the zero hold bonus, the standard-deviation convention and floor, the clip range, the signal phase structure, and the deterministic evaluation mode) are recoverable from the per-task manifests and the archived source files identified there, rather than restated in the frozen file itself. A preceding pilot four-grid study (Section 5.6) used training seeds 42–46; these seeds informed method selection and are excluded from all primary inference. In particular, the method-lock constants (the pressure-prior scale 2.25, the within-intersection normalization with its standard-deviation floor, the clip range, and the zero hold bonus) were selected during the pilot phase and carried unchanged into the frozen confirmatory protocol. This is method selection informed by pilot data; it is compatible with confirmatory hygiene because the confirmatory data did not tune these constants and the confirmatory population consists of ten new independent training seeds 47–56. The frozen exclusion rule permits dropping a seed only for protocol-invalid or corrupted tasks, with the reason reported and the same seed rerun before aggregate results are unblinded; no such exclusion occurred. Ten seeds were frozen as a feasible seed-paired confirmation size, sufficient for an exact all-positive sign outcome to attain the two-sided $p = 0.001953$ reported below, but this was not a formal prospective power calculation. The stopping rule required all 60 planned tasks (3 scenarios × 10 seeds × 2 methods) to run, with no interim efficacy stopping.

2) Paired design and estimand: The training seed is the unit of analysis. Within each training seed and scenario, U is averaged across the five fixed evaluation seeds. These five episodes are nested repeated evaluations of the same trained policy, not independent training replicates. The per-seed effect is the equally weighted mean of log(U_IPPO / U_IPPO-P) across H1, H2, and H3. The aggregate effect is the mean of the ten seed-level log ratios, transformed to percent improvement as $100 \times (1 - \exp(-\text{mean log ratio}))$. The log-ratio is symmetric between methods, and the percent transform directly represents the relative reduction in U. Equal scenario weights were fixed before data collection. H2 and H3 share the same 2×2 network at different demand levels, so two-thirds of the equal scenario weight lies on one topology family. The estimand therefore applies to the frozen scenario set and should not be interpreted as three independent topology replications. Pairing by training seed is the central design choice. Run-to-run variance is a major uncertainty source in deep RL [16], [18], and seed-level pairing removes this variance from the method comparison rather than averaging over it. Reporting seed-level effects also preserves the correct sample size: ten seeds, rather than sixty tasks or three hundred episodes [17].

3) Primary hypothesis test: The pre-registered primary contrast is IPPO-P versus IPPO on U under the estimand of Section 4.2. The test is an exact two-sided paired sign-flip test over the ten seed-level log ratios at $\alpha = 0.05$, with the decision rule "positive aggregate effect and exact two-sided $p < 0.05$". The null hypothesis is symmetry of the seed-level log-ratio distribution about zero. The statistic counts positive seed effects; under this null each seed's effect is equally likely to be positive or negative, so the null distribution is binomial over ten signs and the all-one-direction outcome yields $2 \times 2^{-10} = 0.001953125$ (reported as 0.001953 throughout the remainder of the paper). In statistic form this is the exact paired sign test; the frozen protocol names it a sign-flip test, and under the all-one-direction outcome observed here the sign test and the randomization over the 2^{10} sign assignments of the magnitudes yield the identical p-value, so the label and the reported value agree. The test involves no normal approximation and no variance estimate, which matters at this sample size. One primary contrast is pre-registered; no multiplicity correction applies to it. Exact permutation p-values are reported without zero-floor artifacts [20].

4) Uncertainty quantification: A paired percentile bootstrap over training seeds (20,000 resamples, RNG seed 20261024 [21]) provides 95% confidence intervals. Resampling over training seeds propagates the dominant variance source into the interval and matches the unit-of-analysis choice of Section 4.2. These intervals are descriptive; they do not replace the exact primary test and are not used for decisions. With ten seeds, percentile coverage is imperfect, which is one reason these intervals are labeled descriptive rather than inferential.

5) Secondary analyses: Secondary analyses are: (i) scenario-specific effects on the confirmation seeds, reported descriptively with bootstrap intervals but no per-scenario significance claims; (ii) a pooled 15-seed summary combining pilot and confirmation seeds, labeled descriptive only, since pilot seeds preceded and informed the frozen protocol and do not constitute independent replication of it; (iii) operational audits: completion rates under both denominators, teleport counts, and demand conservation. The secondary service metrics logged under Section 3.6 are not part of these analyses. No secondary outcome was designated as pivotal, no scenario-specific test is treated as confirmatory, and none of the secondary analyses can promote the result beyond the primary contrast's confirmatory scope.

6) Exploratory extensions: Three scale-up settings on the 3×3 grid were run after the confirmatory result was known: H1HIGH (18 route flows × 150 veh/h = 2700 veh/h total; n = 5 training seeds 62–66), H1SAT (a repository label carried for continuity, not a saturation finding; 18 × 300 = 5400 veh/h; n = 5, seeds 67–71), and H1PROBE700 (18 × 350 = 6300 veh/h; n = 2, seeds 72–73). All three are post-confirmation exploratory extensions: the frozen confirmation protocol was copied, not re-registered; none is merged into the primary test; none carries confirmatory status. For the probe, `decision_tree_locked.json` was internally locked at 2026-10-27 10:35:00+08:00 and records `locked_before_results=true`. This lock was author-controlled rather than externally registered. The branch decision rules were locked during the runs, before any evaluation output was available, with n = 2 explicitly incapable of confirming any direction. Flow totals are counted from the 18 route-flow entries; nominal per-intersection demands of 300, 600, and 700 veh/h assume the nine signalized intersections as the denominator. Route-file comments contain inconsistent per-intersection figures, so labels such as "saturated" are not treated as evidence of network saturation. The 18 flows comprise 12 through routes and 6 turning routes, not an allocation balanced across movement classes. The probe additionally carried a pre-locked secondary guard: evidence of completion collapse or teleporting in either method would be treated as a protocol anomaly to be resolved before any mechanism interpretation. No guard condition was triggered.

7) Reproducibility: Task manifests (each task's `protocol_state.json` record) retain SHA-256 hashes of all source files, route-file bindings, variant configurations, checkpoint paths, and the deterministic fresh-environment evaluation mode; source-hash sets are consistent within each scenario across the 60 tasks. The manifest hashes cover the code, configuration, network, and route files that define each task, so a task can be re-derived from the manifest plus the hashed sources. After the frozen runs, the source file `paper_protocol.py` was edited; because the experimental source project was not under recoverable version control at freeze time, its current hash no longer matches the hashes recorded in the frozen manifests, while the other hashed components (environment, agents, scenario files) are unchanged. Bit-exact re-evaluations from stored checkpoints under the current code reproduce the stored per-episode metrics for each re-run task (three tasks across two audits: the two H2 pilot re-checks of Section 5.6 and the censoring-decomposition re-run of Section 6.6; deposit artifacts in the Data Availability statement), supporting functional equivalence of the evaluation path; the drift itself is disclosed rather than resolved. Training writes a checkpoint at episode completion, and evaluation reloads the checkpoint recorded in the manifest. Two classes of provenance were initially not captured at freeze time. Runtime and library versions and hardware descriptions were subsequently recovered from the training machine and are deposited with the archival package's environment snapshot: SUMO binary v1_25_0+0882 with TraCI/sumolib Python client packages 1.27.1, Python 3.9.25, PyTorch 2.8.0+cu126, dual Intel Xeon Gold 6230R with one NVIDIA RTX A6000. SHA-256 hashes of the checkpoint binaries were computed post hoc from the retained files and are deposited alongside. Historical Git commit identifiers for the frozen experimental source remain unrecoverable, and this is disclosed as a known gap (Section 6.6). The protocol specification depends on the frozen file, per-task manifests, and archived hashed sources; the frozen file alone is not self-contained, and post-freeze `paper_protocol.py` drift cannot be reconstructed.

E. Results

1) Protocol completion and integrity checks: All 60 planned tasks were completed. Across the 300 evaluation episodes (60 tasks × 5 evaluation seeds), the demand-state bookkeeping check passed in 300/300 episodes. This check is an identity satisfied by construction (Section 3.6). No teleport events occurred, satisfying the substantive integrity gate. Planned completion ranged from 98.681% to 99.056%. In these runs, completion_rate and planned_completion_rate coincide numerically but retain distinct denominators. The former is completed vehicles over departed vehicles, whereas the latter is normal compliant arrivals over planned demand. They are therefore reported separately throughout. The 300 episodes are repeated evaluations nested within 60 training tasks (10 training seeds × 3 scenarios × 2 methods), not independent training replicates. All inferential statements below use the training seed as the unit of analysis.

Table 1. Scenario specification. Episodes: 50 training episodes per task; 5 deterministic evaluation episodes per task in fresh environments (evaluation seeds 1042–1046).

Scenario	Network	Signalized intersections	Route flows	Total demand (veh/h)	Planned vehicles per evaluation episode	Nominal per-intersection demand (veh/h)	Training seeds (primary)
H1	3×3 grid (`3x3.net.xml`)	9	18 (12 straight, 6 turning)	1800	5004	200	47–56
H2	2×2 grid (`2x2_oversaturated.net.xml`)	4	12 (8 straight, 4 turning)	2400	6672	600	47–56
H3	2×2 grid (`2x2_oversaturated.net.xml`)	4	12 (8 straight, 4 turning)	3000	8340	750	47–56

Notes: Totals are sums over route-flow entries of the route files; the nominal per-intersection demand divides the total by the number of signalized intersections and is an accounting quantity, not a measured saturation level. Network file names, including `2x2_oversaturated.net.xml`, are repository labels and are not treated as evidence of network saturation. Each signalized intersection runs an 8-phase signal program (4 green phases of 33 s plus 4 yellow transitions of 2 s); the controllable action space is the 4 green phases, the yellow interval being an automatic transition. Pilot training seeds 42–46 used the same three scenarios (Section 5.6). Planned-vehicle counts are observed per-episode values from the evaluation runs.

2) Primary confirmatory result: **Table 2. Confirmatory protocol and primary result.**

Item	Value
Estimand	IPPO-P vs IPPO on demand service time U (lower is better); per training seed: mean U over 5 evaluation seeds per scenario, then equal-weight mean log ratio over H1/H2/H3
Population	Ten new independent training seeds 47–56
Effect (percent transform)	**+1.712%**
Bootstrap 95% CI (descriptive)	[1.252%, 2.202%] (20,000 paired resamples, seed 20261024)
Seed direction	10/10 training-seed effects positive
Primary test	Exact two-sided paired sign-flip test whose statistic counts positive seed effects; its equivalence to the paired sign test is specific to the observed all-one-direction outcome; $\alpha = 0.05$, one internally pre-registered contrast, no multiplicity correction
p-value	**p = 0.001953**
Decision	Pre-registered hypothesis confirmed within the frozen scope

The ten seed-level log ratios range from 0.0050 to 0.0323 and are all positive (Figure 2), with a sample SD of 0.00825. The sign-flip p-value is the smallest attainable with ten seeds. It arises from the unanimous direction, not from a large margin in any single seed. The smallest seed effect, 0.0050 for seed 52, would be indistinguishable from noise in isolation. This is the intended interpretation of a paired-seed sign test: the evidence lies in directional consistency across independent training runs, while the bootstrap interval describes magnitude. Descriptive scenario means of U across the confirmation seeds (n = 10 per scenario) are 135.15 s for IPPO versus 134.60 s for IPPO-P in H1, 116.52 s versus 115.33 s in H2, and 123.27 s versus 118.73 s in H3. The effect is modest in magnitude and uniform in direction.

3) Evaluation-realization sensitivity (post hoc reanalysis): All ten training seeds are evaluated on the same five evaluation seeds (1042–1046), so the five episodes are shared realizations across training seeds. The primary pairing and test are unaffected because the training seed remains the unit of analysis. However, the unanimity of the primary result could, in principle, depend on averaging across these shared realizations. A post hoc reanalysis therefore examines direction at the level of individual evaluation episodes using the retained per-episode metrics (deposit: `results/eval_realization_sensitivity/`).

The episode-to-seed mapping is established from run artifacts rather than from the current source tree, whose post-freeze drift is disclosed in Section 4.7: for each of the 60 tasks, the per-episode rewards recorded in the run logs match the stored per-episode metric arrays in order (60/60 tasks). As a recomputation check, the five-episode means reproduce the stored per-task U exactly (maximum absolute difference 0), and the per-scenario and aggregate log ratios reproduce the stored values exactly.

For each evaluation episode e and training seed s, define the single-episode log ratio r(s, e) as the equal-weight mean over H1–H3 of log(U_IPPO(s, e) / U_IPPO-P(s, e)). An exact two-sided sign test over the ten training seeds within each episode gives:

Evaluation episode (seed)	Positive seeds	Exact two-sided p	Mean r(s, e)
e0 (1042)	10/10	0.001953	+0.016638
e1 (1043)	10/10	0.001953	+0.019539

Evaluation episode (seed)	Positive seeds	Exact two-sided p	Mean r(s, e)
e2 (1044)	10/10	0.001953	+0.016289
e3 (1045)	10/10	0.001953	+0.014825
e4 (1046)	9/10	0.021484	+0.018879

The direction is unanimous within four of the five evaluation realizations and 9/10 in the fifth; the single negative seed-episode is training seed 52 at evaluation seed 1046. Realization-level weakness exists at the scenario level (in H1 at evaluation seed 1042 only 3/10 training seeds are positive, p = 0.344, and H2 at 1045 splits 5/10), while H3 is positive in 9–10/10 seeds in every realization. The mean of the single-episode statistics (0.017234) lies 3.4×10^{-5} below the frozen primary statistic (0.017267), the Jensen gap between mean-of-logs and log-of-mean, so this reanalysis quantifies the sensitivity of the reported unanimity to realization sharing without revising the primary estimand. The five within-episode tests share training runs and scenario draws; they are descriptive decompositions, not independent replications or five confirmations.

4) Scenario-specific secondary results: **Table 3. Scenario-specific secondary effects on confirmation seeds (n = 10 per scenario; descriptive, not three independent confirmatory tests).**

Scenario	Effect (percent)	Mean log ratio	Bootstrap 95% CI (descriptive)	Positive seeds	U: IPPO (s)	U: IPPO-P (s)
H1	+0.407%	0.00408	[−0.048, 0.784]	8/10	135.1523	134.6031
H2	+1.023%	0.01028	[0.581, 1.560]	10/10	116.5230	115.3285
H3	+3.675%	0.03744	[2.845, 4.744]	10/10	123.2698	118.7265

These effects are secondary descriptions of the same confirmatory data, not separate hypothesis tests. No per-scenario significance claim is made, and the H1 interval crossing zero is reported as observed. The variation across scenarios, from nearly flat in H1 to largest in H3, motivates the boundary analysis in Section 5.5 rather than a load-response interpretation.

5) 5.5 Exploratory 3×3 scale-up and boundary results: **Table 4. Exploratory 3×3 results (all post-confirmation, exploratory; never merged with the primary test).**

Setting	Total demand (18 flows)	Nominal per intersection	n (training seeds)	Effect (percent)	Positive seeds	Exact sign-flip p (not interpreted)	Mean U: IPPO (s)	Mean U: IPPO-P (s)
H1HIGH	18 × 150 = 2700 veh/h	300 veh/h	5 (62–66)	+0.458%	5/5	0.0625	136.45	135.82
H1SAT	18 × 300 = 5400 veh/h	600 veh/h	5 (67–71)	−0.997%	0/5	0.0625	140.82	142.23
H1PROBE700	18 × 350 = 6300 veh/h	700 veh/h	2 (72–73)	+0.532% / +0.027% (per seed)	2/2	not applicable (descriptive probe)	144.72 / 143.08	143.95 / 143.04

Notes: All rows are post-confirmation exploratory extensions, as stated in the caption. Per-seed values for H1PROBE700 are listed in seed order (72, 73), separated by slashes. The exact p-values of 0.0625 are the minimum attainable with n = 5 and are shown for completeness only; as exploratory quantities they are not interpreted as significance evidence.

Operational checks: in H1HIGH and H1SAT all evaluation episodes passed demand conservation with zero teleports, and the minimum completion rate over episodes was ≥ 0.9864 (H1HIGH) and ≥ 0.9848 (H1SAT) for both methods. In H1PROBE700, conservation passed, teleports were zero, and minimum completion rates were ≥ 0.9848 for both methods.

At 2700 veh/h, the direction matches the confirmatory result (+0.458%, 5/5 positive, exact p = 0.0625, the minimum attainable with n = 5; descriptive only). At 5400 veh/h, the effect reverses to −0.997%, with 0/5 seeds positive. This relative difference occurs under globally degraded conditions rather than indicating method-specific operational failure. Mean U increases by approximately 6–8 s for both methods relative to the 1800 veh/h confirmatory H1 scenario, while completion remains high and no teleports occur. The reversal therefore reflects a change in the prior's relative value under load rather than a breakdown of either controller. At 6300 veh/h, both seeds show paired improvement, but one effect is only +0.027%, a near-tie, and n = 2 cannot confirm any direction.

These points do not support a smooth or monotonic pressure-response curve. Topology, route composition, load, and sample size differ across settings. Flow totals are verified from the route entries (18 × 150/300/350 veh/h), nominal per-intersection figures assume nine signalized intersections, and route-file labels are not treated as evidence of saturation. The substantive result is a negative point and a near-null point alongside the confirmatory positives. The confirmatory gain therefore does not generalize unconditionally across topology-load settings.

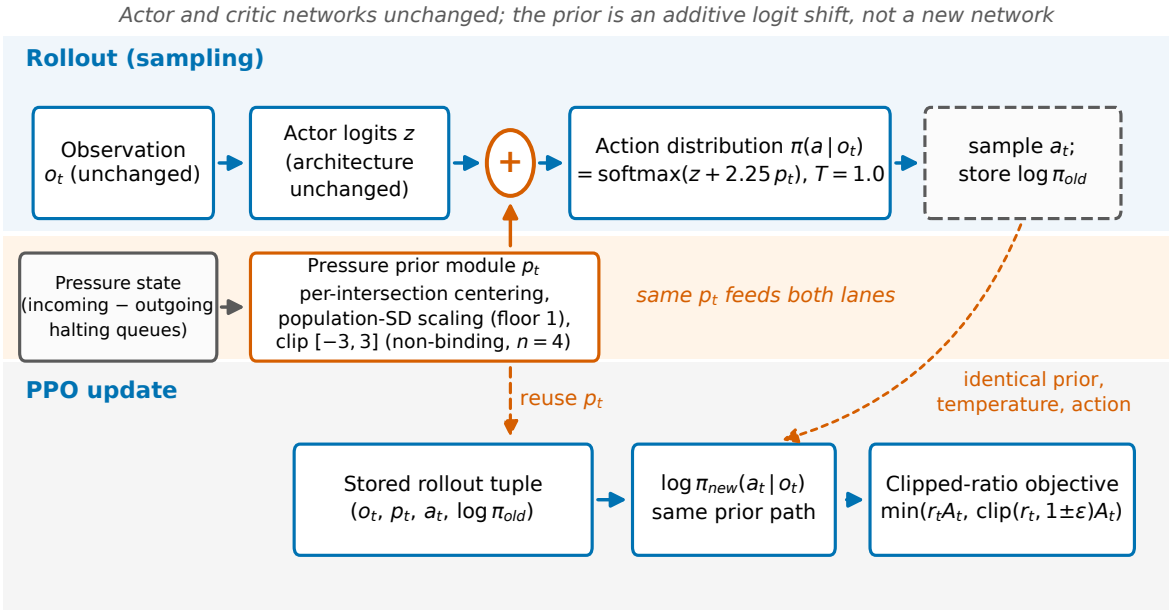


Fig. 1. **Figure 1. Method schematic.** IPPO-P computational and training-consistency diagram: observation → baseline actor logits; pressure state → per-intersection normalized and clipped prior (centered, SD-scaled with population SD and floor 1, clipped to [−3, 3], a bound that never binds at the 4-phase intersections); scaled addition (× 2.25) before the categorical distribution at temperature 1.0; the same stored prior path used for rollout and PPO update, while the update evaluates the stored action under the current actor parameters within the same prior-augmented distribution family. The figure makes explicit that the actor/critic architecture is unchanged: the prior is an additive logit shift, not a new network. *Data source:* Section 3.3–3.4 and the frozen method lock in `protocol_frozen.json`.

6) Pilot four-grid results (exploratory architecture context): The pilot study (training seeds 42–46, same H1/H2/H3 scenarios) preceded the frozen protocol and informed method selection. Its "four-grid" design is the 2 × 2 of architecture (independent PPO vs. a graph-communication variant, `GC`) × prior (absent vs. present); the two GC cells are identical to each other in every parameter except the prior scale (0.0 vs. 2.25), and the GC architecture as a whole changes multiple components simultaneously relative to IPPO (graph-residual independent actor, pressure observation, an action-value head, and soft-Q execution). The five pilot contrasts were: (i) the prior within IPPO (IPPO-P vs IPPO), +2.219% (5/5 positive, exact p = 0.0625); (ii) the prior within the GC architecture (GC-P vs GC-noP), +0.272% (3/5 positive, p = 0.5625); (iii) GC-noP vs IPPO (architecture effect without prior), +1.754% (5/5, p = 0.0625); (iv) GC-P vs IPPO (architecture effect with prior), +2.021% (5/5, p = 0.0625); and (v) the architecture-by-prior interaction, −1.991% (0/5 positive). After Holm correction across the five contrasts, none was significant (minimum Holm-adjusted p = 0.3125). Contrast (i), obtained before the confirmatory protocol was frozen, pointed in the same direction as the later confirmatory contrast and motivated the confirmation. Because the GC variants alter several components at once, no claim, positive or negative, is made about the independent value of graph communication. Pilot data never enter confirmatory inference; the pooled 15-seed effect of +1.881% (15/15 positive) is reported as descriptive only. A post hoc forensic recheck of the one anomaly flagged by a consistency audit (bit-identical five-episode evaluation vectors for the H2 pilot seeds 45 and 46) confirmed two independently trained checkpoints (distinct weights and distinct training curves) whose deterministic policies coincide on the shared evaluation trajectories; the pilot values stand as computed (deposit: `results/audit_recheck_v1/`).

7) External descriptive anchors (post hoc, not part of the confirmatory protocol): After the confirmatory analysis was complete, two non-learning anchors were run under the same evaluation setup. The nominal embedded fixed-time program is 140 s (4 × 33 s green + 4 × 2 s yellow). The harness requests the successor green at each 10 s decision slot, while the environment enforces the 33 s minimum green and yellow transition, producing realized greens of 33–43 s and realized cycles of approximately 156–172 s. Max-Pressure-Descriptive is a constrained descriptive variant that, at each decision step, extends the phase with the largest total incoming-minus-outgoing halting-count proxy across controlled movements (minimum green 33 s; ties broken by program order). It is not a canonical unconstrained max-pressure implementation. Both use the identical metrics path, demand window, and evaluation seeds 1042–1046, with no training. These anchors were constructed after the confirmatory analysis; they enter no statistical test, were not part of the pre-registered design, and do not alter any confirmatory claim.

Table 5. External descriptive anchors (n = 5 evaluation episodes per cell; post hoc, descriptive only).

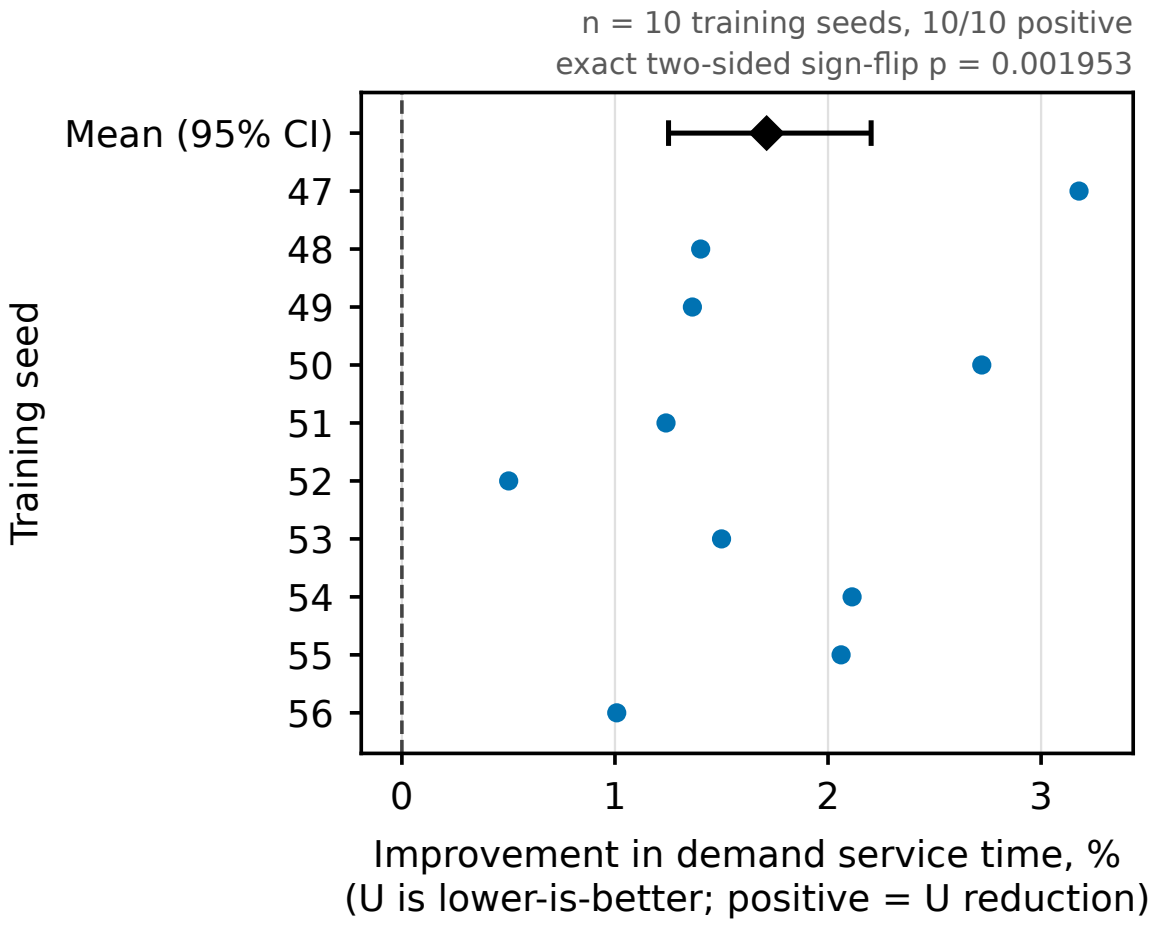


Fig. 2. **Figure 2. Seed-level effects.** Paired log-ratio effects of IPPO-P vs IPPO per training seed (47–56), zero reference line, aggregate estimate with descriptive bootstrap interval; n = 10 training seeds; exact two-sided p = 0.001953. The plotted quantities are raw log ratios; the percent transform of a log ratio r is $100 \times (1 - \exp(-r))$, under which the aggregate point corresponds to +1.712%. *Data source:* `seed_log_ratios` in `confirmation_summary.json` (0.0323, 0.0141, 0.0137, 0.0276, 0.0125, 0.0050, 0.0151, 0.0214, 0.0208, 0.0101 for seeds 47–56).

U SD is the sample standard deviation over the five evaluation episodes (ddof = 1); mean U for the learned arms is reported in Table 3.

Scenario	Controller	U (s, mean)	U (SD)	Planned completion (min)	Teleports	Conservation
H1	Fixed-Time	224.87	5.43	0.978	0	pass
H1	Max-Pressure	144.63	2.19	0.987	0	pass
H2	Fixed-Time	223.79	28.78	0.968	0	pass
H2	Max-Pressure	121.38	1.83	0.988	0	pass
H3	Fixed-Time	799.62	11.58	0.853	0	pass
H3	Max-Pressure	122.21	0.62	0.987	0	pass

Both learned policies have substantially lower U than the fixed-time anchor in every scenario. Relative to the constrained descriptive max-pressure anchor, which uses a 33 s minimum green and a halting-count proxy rather than a canonical unconstrained implementation, IPPO-P has a lower descriptive mean U in all three scenarios (134.60 vs 144.63; 115.33 vs 121.38; 118.73 vs 122.21). These differences correspond to 6.9%, 5.0%, and 2.9% of the anchor's U. IPPO has a lower mean U in H1 and H2 but a higher mean U in H3 (123.27 vs 122.21). Because each cell contains five evaluation episodes and the anchors were constructed post hoc, these comparisons provide scale context rather than ranked claims. The fixed-time breakdown in H3 (planned completion 85.3%, U ≈ 800 s) reflects an unadaptive program under the heaviest demand. It is reported as anchor behavior, not as an algorithm comparison.

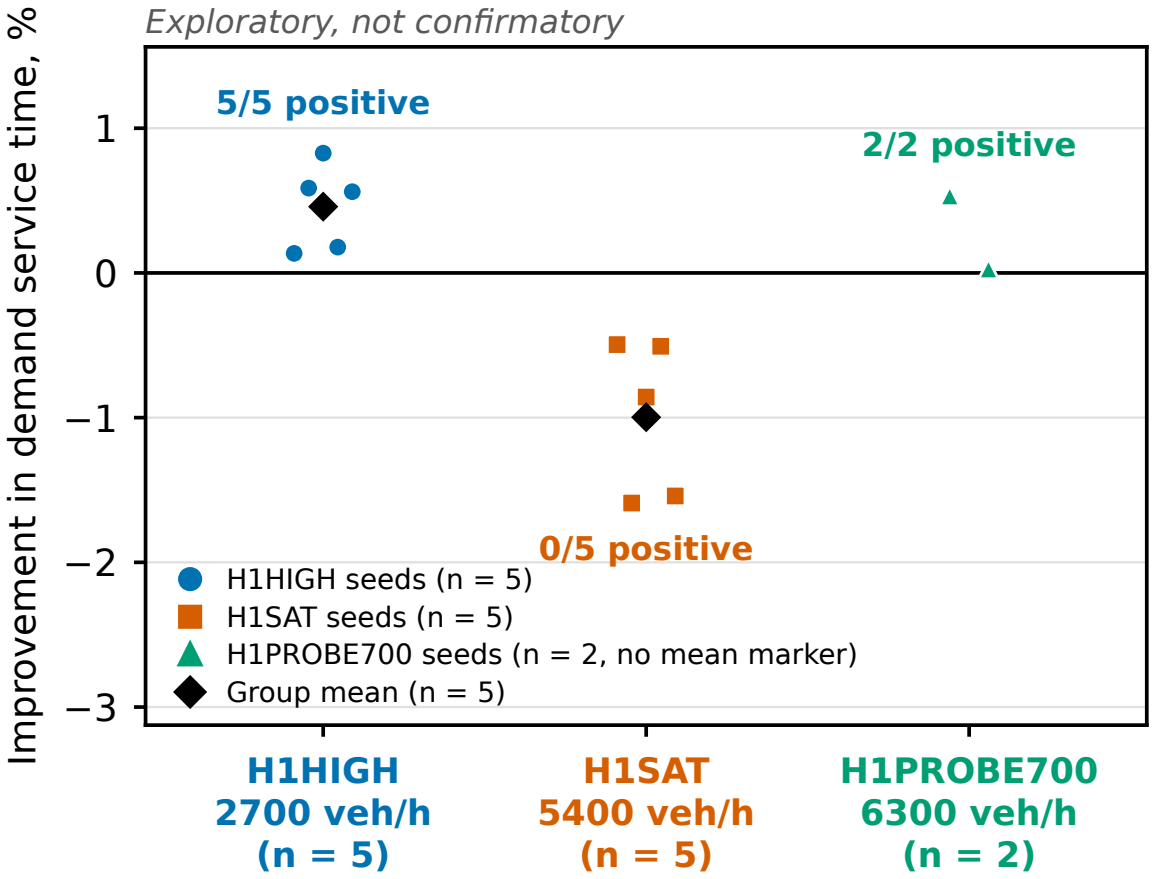


Fig. 3. **Figure 3. Exploratory boundary results.** Discrete effect points for H1HIGH (n = 5), H1SAT (n = 5), and H1PROBE700 (per-seed points, n = 2), with sample-size labels; the legend identifies the probe as per-seed points shown without a mean marker; scenario and demand annotated per point; no smoothed connecting line and no pooled inferential test. *Data source:* `scaleup_summary.json` (H1HIGH), `scaleup_v2_summary.json` (H1SAT), `decision_tree_locked.json` and per-task evaluation metrics (H1PROBE700).

F. Discussion

1) Main finding: Under a protocol frozen before data collection, training-consistent integration of a normalized pressure prior into the PPO policy logits improved demand service time relative to a matched IPPO baseline. The effect was +1.712%, with all ten training-seed effects positive and an exact two-sided p = 0.001953, the smallest attainable with ten seeds. A post hoc decomposition by evaluation episode (Section 5.3) retained unanimous direction in four of the five shared evaluation realizations and 9/10 in the fifth. This pattern is consistent with the aggregate unanimity not depending solely on averaging across shared realizations. In absolute terms, the descriptive scenario means imply descriptive per-vehicle savings of approximately 0.5 s in H1, 1.2 s in H2, and 4.5 s in H3. The H3 estimate applies across approximately 8,340 planned vehicles per evaluation episode. The remaining secondary service metrics, including completed-vehicle travel time and delay, lane-aggregated queue, waiting, and speed, were logged but excluded from the pre-registered secondary analyses (Section 3.6). They are not reported, their directions are not adjudicated, and they provide neither descriptive nor confirmatory evidence here. The finding is confirmatory only within the frozen scenarios, metric, seeds, and analysis. It is not a general claim about traffic signal control.

2) Scope of the evaluated treatment: The evaluated treatment is the complete training-consistent prior integration. Through stored-prior reuse, the same modified distribution governs rollout sampling and PPO likelihood evaluation, preserving importance-ratio semantics (Section 3.4). This scope excludes two weaker interpretations. First, the result does not show that biasing a trained IPPO policy toward high-pressure phases only during evaluation would help. Such a post hoc intervention was not tested and would not be equivalent. Second, the result does not show that pressure belongs in the reward. The reward was identical across methods, so the comparison separates the complete prior-integration treatment from reward changes. However, IPPO-P also includes an extra halting-count channel, and the design does not isolate that channel's effect from the effect of adding its derived prior to the logits. Thus, IPPO-P is not an isolated logit-arithmetic treatment. The scale (2.25) and normalization were selected during the pilot phase

and fixed unchanged in the method lock (Section 4.1). No tuning over these values used the confirmatory data.

3) Boundary evidence: The exploratory 3×3 results bound the claim. At a nominal per-intersection demand of 600 veh/h, the effect is positive on the 2×2 grid (H2, +1.023%, confirmation seeds) but negative on the 3×3 grid (H1SAT, −0.997%, 0/5 seeds). Within the 3×3 family, the effect changes from +0.41% at 200 veh/h (H1; confirmatory, n = 10) to +0.46% at 300 (H1HIGH; exploratory, n = 5), −1.00% at 600 (H1SAT; exploratory, n = 5), and a near-tie at 700 (H1PROBE700; exploratory probe, n = 2). These settings differ simultaneously in topology, route composition, and load. The topology-load contrast is therefore confounded, and no factor attribution is attempted. The 12-through/6-turn composition of the 18 flows is constant across the 3×3 settings, but this constancy does not balance the load ladder by movement class. The data indicate that the confirmatory gain is conditional. It can persist under increased load on one topology, reverse on another, and approach a near-tie at the highest tested load. The near-tie warrants particular caution. At 6300 veh/h, the two seeds point in the same direction, but their margins (+0.532% and +0.027%) differ by an order of magnitude. Even the direction is therefore not firmly established at this load. The H1SAT result contradicts universal-superiority claims, and none are made.

4) Mechanism: a hypothesis, not a finding: A variance-stabilization account remains plausible but untested. Under this account, pressure biases sampling toward queue-balancing phases and reduces destructive action variance during training. The recorded data contain scenario-level aggregates rather than intersection-level traces of pressure, chosen actions, or phase switches. Without those traces, no causal mechanism can be identified, and none is claimed. The H1SAT degradation pattern, in which both methods have worse absolute U but IPPO-P is relatively worse, and the near-tie at 6300 veh/h are consistent with competing bias-cost and stabilization-benefit terms. Their balance may shift with operating conditions, but this interpretation remains a hypothesis for future instrumented experiments. Two observable consequences could help distinguish it. The first is seed-to-seed dispersion of U under the prior in regimes with a positive mean effect. The second is intersection-level decision traces showing whether the prior disproportionately corrects queue-imbalanced decision slots. Such traces were not recorded in the reported runs.

5) Relation to prior work: Reported pressure-informed RL controllers place pressure in the reward [9], [10] or state [11], largely within DQN-family value-based agents. The present work holds the reward and actor observation vector fixed. It computes a pressure signal through a separate halting-count channel and adds the resulting prior to the logits of an on-policy PPO policy during both rollout and update. This logit-versus-state distinction is structural and was not experimentally isolated here. In an on-policy method, the sampling distribution is intrinsic to the learning rule, making the policy logits a natural integration point for such a prior. No head-to-head comparison with DQN-family pressure methods was conducted. Accordingly, no claim is made that PPO with a prior outperforms DQN with a pressure-based reward. The claims concern the prior's effect on PPO under matched conditions. The contribution is therefore an evaluation of treatment consistency under an internally pre-registered protocol with seed-level reporting, rather than a new traffic-pressure principle or architectural advance. Classical max-pressure control [8] and its queueing-theoretic origins [7] provide the conceptual basis for the prior. The only performance comparison is the explicitly post hoc descriptive anchor in Section 5.7, from which no ranked or inferential claim is drawn.

6) Limitations:

- One simulator family (SUMO) and synthetic grid configurations only; no real-world or field data.
- The external anchors (fixed-time program; descriptive max-pressure) were constructed after the confirmatory analysis and are reported as descriptive context only (Section 5.7); they follow documented fixed rules but were not part of the pre-registered design. The max-pressure anchor is a constrained descriptive variant with a 33 s minimum green and a halting-count proxy, not a canonical unconstrained implementation.
- Ten confirmatory training seeds; exploratory extensions use n = 5 and n = 2, the latter incapable of confirming any direction. A single final checkpoint after 50 training episodes and one prior scale cannot establish robustness to training budget, checkpoint choice, or scale.
- Topology and demand are not factorially separated in the 3×3 extensions; the H2-versus-H1SAT comparison is confounded.
- The contribution of censored (incomplete) vehicles to the primary metric U is quantified by an archived decomposition on one representative confirmation task (H1 under IPPO, training seed 47; deposit `results/censoring_decomposition_v1/`): a bit-exact re-run of its five evaluation episodes, cross-checked against the simulator's per-vehicle tripinfo records, places the censored share of U mass at 0.60%–0.73% per episode (0.67% pooled over the five episodes), with censored vehicles 1.22% of planned demand (304 of 25,020), all of them still in the network at horizon end, and zero teleports. This is a single-task benchmark for one arm and scenario, not a per-scenario or matched-arm measurement; on it, normalizing over completed vehicles only (rather than censoring) would yield 136.69 s rather than 135.94 s. The censoring contribution is therefore small on the evidence available, but its arm and scenario scope remains limited.
- No intersection-level pressure, action, or phase-switch traces for the 5400/6300 veh/h runs, precluding mechanism identification. No effective-decision-slot fraction was archived, so sample size by executable decision opportunity cannot be quantified.

- Pilot results informed method selection and remain separate from confirmatory inference; the pooled 15-seed summary is descriptive only. All hyperparameters, including the prior's constants, were carried over from the same-scenario pilot, so the confirmatory design removes confirmatory-data tuning but not the residual optimism that pilot-informed magnitudes carry (Sections 4.1, 6.2).
- The signal timing settings (a 2 s yellow interval, shorter than the 3–5 s typical of urban geometry, and a 10 s decision interval) are fixed and shared by both methods, but they were not varied. Sensitivity to finer timing and to the 10 s grid is not quantified, and the prior's interaction with different timing regimes is untested.
- Runtime/library versions, hardware descriptions, and checkpoint-file hashes were not captured in the frozen manifest; the first two have since been recovered from the training machine and deposited with the archival package's environment snapshot (SUMO v1_25_0+0882 binary, TraCI/sumolib client packages 1.27.1, Python 3.9.25, PyTorch 2.8.0+cu126, dual Xeon Gold 6230R + RTX A6000), and checkpoint-file SHA-256 hashes were computed post hoc from the retained files. The SUMO binary and Python clients came from separate installed distributions, and their compatibility pairing was recovered post hoc rather than locked. Historical Git commit identifiers for the frozen experimental source remain unavailable. The protocol lock is not self-contained, and post-freeze drift in the current `paper_protocol.py` cannot be reconstructed because the source project lacked recoverable version control. Bit-exact checkpoint re-evaluations under the current code reproduce the stored metrics for each re-run task (Section 4.7), supporting functional equivalence of the evaluation path without resolving this provenance gap.
- The earliest-task timestamp supports temporal ordering between the internal freeze and data generation, but both records remain author-controlled rather than externally registered.
- No evidence of universal superiority, no real-world deployment, and no safety guarantees of any kind are implied.

7) Implications for future work: Two extensions follow directly as future pre-registered work rather than additional post hoc experiments. The first is a factorial topology-by-load design that separates the confounds in the 3×3 boundary results. The second is local trace logging, including per-intersection pressure, sampled actions, and phase switches, sufficient to test the variance-stabilization hypothesis. A matched arm should supply halting counts as actor state without a logit prior, isolating the structural logit-versus-state distinction. A further comparison would add the same prior to a MAPPO arm, testing whether the effect persists with a centralized critic [14]. A third extension is a training-budget and intermediate-checkpoint study, paired with a sensitivity protocol for the prior's two free constants, the scale 2.25 and the clip range. These constants should be evaluated as a pre-registered family of contrasts rather than tuned post hoc; the confirmatory result covers only the frozen values. Increasing the seed count beyond ten would narrow the descriptive intervals and support multiplicity-aware analysis of scenario-specific contrasts. Matched censoring decompositions across arms and scenarios, together with finer timing and decision-grid sensitivity, should quantify the remaining metric and control-timing boundaries. For practitioners, the operational implication is procedural. The prior's implemented cost is one addition per logit per decision step, an analytical characterization because no wall-clock or memory overhead was measured. It does not change the deployed network architecture. Adoption should therefore rely on the same seed-paired, frozen-protocol evaluation used here rather than on any single-run pilot, including those reported in this paper.

G. Conclusion

Under an internally pre-registered, hash-locked protocol frozen before data collection, this IPPO-internal controlled evaluation found that a normalized, clipped pressure prior reduced demand service time relative to a matched IPPO baseline. The prior was added to the independent PPO policy logits identically during rollout and update. Across three synthetic grid scenarios spanning two topology families, the effect was +1.712% (95% bootstrap CI [1.252%, 2.202%], descriptive), with 10/10 training-seed effects positive and an exact two-sided paired sign-flip $p = 0.001953$. Because H2 and H3 share the 2×2 network, equal scenario weighting assigns two-thirds of the aggregate estimand to that topology family. The intervention leaves the reward, actor observation vector, architecture, and training budget unchanged. However, it adds a halting-count pressure channel whose derived prior is incorporated into the logits and kept consistent between the behavior policy and updated likelihood. This conclusion is confined to the frozen H1/H2/H3 scope and this specific training-consistent prior-integration treatment. It compares IPPO-P with IPPO alone and implies nothing about performance relative to external controllers or published benchmarks.

Post-confirmation exploratory evidence cautions against universal extrapolation. The intervention showed a positive descriptive effect at 2700 veh/h on the 3×3 grid, a consistent negative effect at 5400 veh/h, and an inconclusive near-tie at 6300 veh/h. No pressure threshold is established, no variance-stabilization mechanism is proven, and no general superiority over existing traffic-control approaches is claimed. The confirmatory evidence covers one point in the prior's design space: scale 2.25 at temperature 1.0. Sensitivity to these constants remains untested. For scale context, both learned policies have substantially lower U than the fixed-time anchor in every scenario. IPPO-P's descriptive mean U is 2.9%–6.9% lower than that of the constrained descriptive max-pressure anchor, which uses a 33 s minimum

green and halting-count proxy rather than a canonical unconstrained implementation and in turn edges IPPO in the heaviest scenario. These figures provide scale context rather than ranked claims. The intervention is reported exactly as evaluated: one frozen method, one metric, ten seeds, and three scenarios. Its boundaries are reported with equal precision. The practical conclusion is narrow. Training-consistent pressure priors are an architecturally minimal modification, requiring one addition per logit at decision steps without changing the deployed network. At the frozen 50-episode budget, they improved the aggregate estimand in every confirmatory seed tested here, but their benefit depends on the operating setting and requires future factorial, instrumented studies.

H. Declarations

1) Data Availability Statement: All data supporting the reported results were generated by the frozen protocol `ippo_prior_confirmation_v1` and its documented exploratory extensions. The archival package contains: the frozen protocol and hash file (`data/results/ippo_prior_confirmation_v1/protocol_frozen.json` and `.sha256`); the confirmation summary with all seed-level effects (`data/results/ippo_prior_confirmation_v1/confirmation_summary.json`); per-task manifests, training logs, and per-evaluation-seed metrics under `data/results/`; scenario network and route files under `code/nets/grid-3x3/` and `code/nets/2x2-grid/`; exploratory outputs under `data/results/ippo_prior_scaleup_v1/`, `ippo_prior_scaleup_v2/`, `ippo_prior_probe_700/`, and `formal_four_grid_v1/`; post-hoc external anchors under `data/results/baselines_descriptive_v1/`, with generators in `code/agents/` and `code/tools/`; and the audit/reanalysis deposits `data/results/audit_recheck_v1/`, `eval_realization_sensitivity/`, and `censoring_decomposition_v1/`, with their generating scripts under `code/tools/`. The frozen method lock records the configuration identifier `pressure_zscore_hold`. Checkpoint binaries are not included in the deposit and no simulator or software binaries are claimed as available. Instead, `checkpoint_hashes.sha256` records SHA-256 hashes for the 288 retained checkpoint and resume-state files. Runtime and library versions and hardware descriptions are recorded under `environment_snapshot/`. Historical Git commit identifiers for the frozen experimental source remain unavailable, although the deposit package itself is versioned for release. Package publication is pending, and independent access is not yet available until the DOI below is filled. [**FILL: public archival DOI/repository (e.g., Zenodo) — add upon deposit or acceptance.**]

2) Ethics Declaration: This study uses only synthetic microscopic traffic simulation. It involved no human participants, human data, or animals, and therefore required no ethics-committee approval.

3) Author Contributions (CRediT): **Rongtao Zhu:** Conceptualization, Methodology, Software, Investigation, Data Curation, Formal Analysis, Visualization, Writing – Original Draft. **Zhaoxia He:** Methodology, Validation, Formal Analysis, Writing – Review & Editing. **An Zhao:** Software, Investigation, Data Curation. **Shubin Wei:** Investigation, Visualization, Writing – Review & Editing. **Qiran Li:** Investigation, Data Curation, Writing – Review & Editing.

All authors read and approved the final manuscript and take full responsibility for its contents.

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6) Use of Artificial Intelligence: Artificial-intelligence tools assisted with data aggregation, statistical checking, and manuscript drafting under direct human supervision, following the frozen protocol and its pre-specified analyses. Every quantitative result is traceable to the frozen protocol outputs listed in the Data Availability Statement; no AI-generated numbers were introduced. The authors reviewed, verified, and revised all content and take full responsibility for the manuscript.

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